

Math 170E Notes - W2

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1 W2 (Ch1.1-1.3)

Example 1.1. (PSI 1.1.1) Of a group of patients having injuries, 28% visit both a physical therapist and a chiropractor and 8% visit neither. Say that the probability of visiting a physical therapist exceeds the probability of visiting a chiropractor by 16%. What is the probability of a randomly selected person from this group visiting a physical therapist?

Solution

Problem Setup Let S be the randomly selected person. Define 2 events:

$$A = \{S \text{ visits physical therapist (PT)}\}$$

and

$$B = \{S \text{ visits chiropractor (Ch)}\}$$

We are given $\mathbb{P}((A \cup B)') = 0.08$, $\mathbb{P}(A \cap B) = 0.28$ and $P(A) = P(B) + 0.16$ and we are asked to find $\mathbb{P}(A)$.

Method 1 We can use *Venn Diagrams* again.

Method 2 We can use Principle of Inclusion and exclusion.

Theorem 1.1 (Permutation of Different Objects). Assume we have n different objects: $O_1, O_2, \dots, O_n$.

1. There are exactly $n! := 1 \cdot 2 \cdot 3 \cdots (n-1) \cdot n$ different ways to permute all of n different objects
2. Now assume we want to permute only $r \leq n$ of the objects. Then there exactly $n \cdot (n-1) \cdot (n-2) \cdots (n-r+1) = \frac{n!}{(n-r)!}$ ways to permute them. This is usually denoted by ${}_nP_r = \frac{n!}{(n-r)!}$

Example 1.2. 10 horses are racing in a derby. The first, the second, and the third will be determined and awarded. How many possible outcomes are possible?

Solution

Theorem 1.2 (Combination of Different Objects). Assume we have n different objects: $O_1, O_2, \dots, O_n$. Now assume we want to **choose** only $r \leq n$ of the objects. Then there exactly

$$\frac{n \cdot (n-1) \cdot (n-2) \cdots (n-r+1)}{r!} = \frac{n!}{(n-r)!r!}$$

ways to choose them. This is usually denoted by $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

Example 1.3. We have 10 different flavored candies and we want to choose 3 of them. In how many ways we can do it?

Solution

Example 1.4. (PSI 1.2.5) How many four-letter code words are possible using the letters in IOWA if

1. The letters may not be repeated?
2. The letters may be repeated?

Solution

Example 1.5. (PSI 1.2.5-modified) How many code words up to 5 letters are possible using the letters in IOWA if

1. The letters may not be repeated?
2. The letters may be repeated?

Solution